Exercises: Convex Sets, Convex Functions, and Jensen's Inequality – Answer Key

Mathematical Foundations for Machine Learning

Convex Sets, Cones, and Convex Hulls

Exercise 1.1 [Easy]
Determine which of the following subsets of $\mathbb{R}^n$ are convex. Justify each answer.

(a) $S_1 = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\}$ (the closed unit ball).
(b) $S_2 = \{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$ (the unit circle).
(c) $S_3 = \{x \in \mathbb{R}^n \mid Ax \leq b\}$ for $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ (a polyhedron).

Solution

(a) Convex. Let $x, y \in S_1$ and $\lambda \in [0, 1]$. By the triangle inequality:

$$\|\lambda x + (1-\lambda)y\|_2 \leq \lambda\|x\|_2 + (1-\lambda)\|y\|_2 \leq \lambda \cdot 1 + (1-\lambda) \cdot 1 = 1.$$

So $\lambda x + (1-\lambda)y \in S_1$.
(b) Not convex. Take $x = (1, 0)$ and $y = (-1, 0)$, both on the unit circle. Their midpoint is $\frac{1}{2}x + \frac{1}{2}y = (0, 0)$, which satisfies $0^2 + 0^2 = 0 \neq 1$, so the midpoint is not in S_2.
(c) Convex. Let $x, y \in S_3$, so $Ax \leq b$ and $Ay \leq b$. For $\lambda \in [0, 1]$:

$$A(\lambda x + (1-\lambda)y) = \lambda Ax + (1-\lambda)Ay \leq \lambda b + (1-\lambda)b = b.$$

Alternatively, S_3 is an intersection of half-spaces $\{x \mid a_i^T x \leq b_i\}$, each of which is convex. The intersection of convex sets is convex.

Exercise 1.2 [Easy]
Let $K \subset \mathbb{R}^n$ be defined as $K = \{x \in \mathbb{R}^n \mid x_i \geq 0,\ i = 1, \ldots, n\}$ (the non-negative orthant).

(a) Show that K is a convex cone.
(b) Is K a proper cone? Justify your answer.

Solution

(a) We verify the two properties of a convex cone.
Closed under non-negative scaling: If $x \in K$ and $\alpha \geq 0$, then $(\alpha x)_i = \alpha x_i \geq 0$ for all i, so $\alpha x \in K$.
Closed under addition: If $x, y \in K$, then $(x + y)_i = x_i + y_i \geq 0$ for all i, so $x + y \in K$.
Equivalently, for $\lambda \in [0, 1]$: $\lambda x + (1-\lambda)y$ has all non-negative components, so K is convex.
Combined with the scaling property, K is a convex cone.
(b) A proper cone must be: (i) convex, (ii) closed, (iii) solid (non-empty interior), and (iv) pointed ($K \cap (-K) = \{0\}$).
(i) Shown in (a). (ii) K is closed since each constraint $x_i \geq 0$ defines a closed half-space. (iii) $\text{int}(K) = \{x \mid x_i > 0\}$ is non-empty. (iv) If $x \in K \cap (-K)$, then $x_i \geq 0$ and $-x_i \geq 0$ for all i, so $x = 0$.
Thus K is a proper cone.

Exercise 1.3 [Medium]
Let $C_1, C_2 \subset \mathbb{R}^n$ be convex sets.

(a) Prove that $C_1 \cap C_2$ is convex.

(b) Give a counterexample showing that $C_1 \cup C_2$ need not be convex.

(c) The *Minkowski sum* is defined as $C_1 + C_2 = \{x + y \mid x \in C_1,\ y \in C_2\}$. Prove that $C_1 + C_2$ is convex.

Solution

(a) Let $x, y \in C_1 \cap C_2$ and $\lambda \in [0, 1]$. Since $x, y \in C_1$ and C_1 is convex, $\lambda x + (1 - \lambda)y \in C_1$. Similarly, $\lambda x + (1 - \lambda)y \in C_2$. Therefore $\lambda x + (1 - \lambda)y \in C_1 \cap C_2$.
(b) Take $C_1 = \{(x, 0) \mid x \in [0, 1]\}$ and $C_2 = \{(x, 0) \mid x \in [2, 3]\}$ in $\mathbb{R}^2$. Both are convex (line segments). But $(0.5, 0) \in C_1$ and $(2.5, 0) \in C_2$ with midpoint $(1.5, 0) \notin C_1 \cup C_2$.
(c) Let $z_1, z_2 \in C_1 + C_2$, so $z_1 = x_1 + y_1$ and $z_2 = x_2 + y_2$ with $x_1, x_2 \in C_1$ and $y_1, y_2 \in C_2$. For $\lambda \in [0, 1]$:

$$\lambda z_1 + (1 - \lambda)z_2 = \underbrace{\lambda x_1 + (1 - \lambda)x_2}_{\in C_1} + \underbrace{\lambda y_1 + (1 - \lambda)y_2}_{\in C_2} \in C_1 + C_2.$$

Exercise 1.4 [Medium]
The convex hull of a set $S \subset \mathbb{R}^n$ is defined as

$$\operatorname{conv}(S) = \Big\{ \sum_{i=1}^{k} \lambda_i x_i \;\Big|\; x_i \in S,\ \lambda_i \geq 0,\ \sum_{i=1}^{k} \lambda_i = 1,\ k \in \mathbb{N} \Big\}.$$

(a) Show that $\operatorname{conv}(S)$ is convex.

(b) Show that $\operatorname{conv}(S)$ is the smallest convex set containing S, i.e., if C is any convex set with $S \subseteq C$, then $\operatorname{conv}(S) \subseteq C$.

(c) In the context of machine learning, suppose a neural network's weight space is $\mathbb{R}^d$ and we have a finite set of trained models $w_1, \ldots, w_k \in \mathbb{R}^d$. Describe geometrically what $\operatorname{conv}(\{w_1, \ldots, w_k\})$ represents. If the loss function $L(w)$ is convex, what can you say about the loss at any point in this convex hull?

Solution

(a) Let $u, v \in \text{conv}(S)$, so $u = \sum_{i=1}^{p} \alpha_i x_i$ and $v = \sum_{j=1}^{q} \beta_j y_j$ with $\alpha_i, \beta_j \geq 0$, $\sum \alpha_i = \sum \beta_j = 1$, and $x_i, y_j \in S$. For $\lambda \in [0, 1]$:

$$\lambda u + (1-\lambda)v = \sum_{i=1}^{p} \lambda\alpha_i\, x_i + \sum_{j=1}^{q} (1-\lambda)\beta_j\, y_j.$$

The coefficients $\lambda\alpha_i$ and $(1-\lambda)\beta_j$ are all non-negative and sum to $\lambda \sum \alpha_i + (1-\lambda) \sum \beta_j = 1$. All points $x_i, y_j \in S$, so this is a convex combination of points in S, hence belongs to $\text{conv}(S)$.

(b) Let C be convex with $S \subseteq C$. Take any $z \in \text{conv}(S)$, so $z = \sum_{i=1}^{k} \lambda_i x_i$ with $x_i \in S \subseteq C$. We prove by induction on k that $z \in C$.

Base: $k = 1$: $z = x_1 \in C$.

Step: Assume the result for $k-1$. If $\lambda_k = 1$, then $z = x_k \in C$. Otherwise, write

$$z = (1-\lambda_k) \underbrace{\sum_{i=1}^{k-1} \frac{\lambda_i}{1-\lambda_k} x_i}_{\in C \text{ by induction}} + \lambda_k x_k.$$

Since C is convex and both points are in C, we get $z \in C$.

(c) Geometrically, $\text{conv}(\{w_1, \ldots, w_k\})$ is the polytope (simplex if the points are in general position) spanned by the k model weight vectors. Points inside represent *model ensembles* obtained by interpolating the weights.

If $L(w)$ is convex and $w = \sum_{i=1}^{k} \lambda_i w_i$ with $\lambda_i \geq 0$, $\sum \lambda_i = 1$, then by Jensen's inequality:

$$L(w) = L\Big(\sum \lambda_i w_i\Big) \leq \sum \lambda_i L(w_i) \leq \max_i L(w_i).$$

So the loss at any convex combination is at most the worst individual model loss.

Exercise 1.5 [Hard]

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Define the *sub-level set*

$$L_\alpha(f) = \{x \in \mathbb{R}^n \mid f(x) \leq \alpha\}.$$

(a) Prove that $L_\alpha(f)$ is a convex set for every $\alpha \in \mathbb{R}$.

(b) Give a counterexample showing that the converse is false: a function can have all sub-level sets convex yet fail to be convex.

(c) Let $f_1, \ldots, f_m : \mathbb{R}^n \to \mathbb{R}$ be convex functions and define $g(x) = \max_{i=1,\ldots,m} f_i(x)$. Prove that g is convex. Then show that the feasible set

$$\mathcal{F} = \{x \in \mathbb{R}^n \mid f_i(x) \leq 0,\ i = 1, \ldots, m\}$$

is convex.

(d) In the context of training a neural network with m constraints on fairness metrics, each of the form $f_i(w) \leq 0$ where f_i is convex, explain why the feasible region of model parameters is convex even though the intersection of constraints may be complex.

Solution

(a) Let $x, y \in L_\alpha(f)$, so $f(x) \le \alpha$ and $f(y) \le \alpha$. For $\lambda \in [0,1]$, by convexity of f:

$$f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda) f(y) \le \lambda\alpha + (1-\lambda)\alpha = \alpha.$$

Thus $\lambda x + (1-\lambda)y \in L_\alpha(f)$.
(b) Consider $f : \mathbb{R} \to \mathbb{R}$ defined as $f(x) = -x^2$. Every sub-level set $L_\alpha(f) = \{x \mid -x^2 \le \alpha\}$ is either $\mathbb{R}$ (if $\alpha \ge 0$), or $\{x \mid x^2 \ge -\alpha\} = (-\infty, -\sqrt{-\alpha}] \cup [\sqrt{-\alpha}, \infty)$ (if $\alpha < 0$). Wait—this is not convex. A simpler example: $f(x) = \sqrt{|x|}$. Its sub-level sets $L_\alpha = [-\alpha^2, \alpha^2]$ for $\alpha \ge 0$ are all intervals (convex), but f is not convex since $f(\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot (-1)) = f(0) = 0 < \frac{1}{2}(1+1) = 1$. In fact, a concave function works: $f(x) = -x^2 + 1$. Its sub-level sets: $L_\alpha = \{x : -x^2 + 1 \le \alpha\} = \{x : x^2 \ge 1 - \alpha\}$, which is $\mathbb{R}$ for $\alpha \ge 1$, and for $\alpha < 1$ is $(-\infty, -\sqrt{1-\alpha}] \cup [\sqrt{1-\alpha}, \infty)$, which is not convex. So we need a non-convex function with all convex sub-level sets. Take any monotone function, e.g., $f(x) = -e^x$ on $\mathbb{R}$. Then $L_\alpha = \{x : -e^x \le \alpha\}$. For $\alpha \ge 0$, $L_\alpha = \mathbb{R}$. For $\alpha < 0$: $L_\alpha = \{x : e^x \ge -\alpha\} = [\ln(-\alpha), \infty)$, which is convex. But $f(x) = -e^x$ is concave (since $f'' = -e^x < 0$), hence not convex.
(c) *g is convex:* For $\lambda \in [0,1]$ and any $x, y \in \mathbb{R}^n$:

$$g(\lambda x + (1-\lambda)y) = \max_i f_i(\lambda x + (1-\lambda)y) \le \max_i \left[\lambda f_i(x) + (1-\lambda) f_i(y)\right]$$

$$\le \lambda \max_i f_i(x) + (1-\lambda) \max_i f_i(y) = \lambda g(x) + (1-\lambda) g(y).$$

$\mathcal{F}$ is convex: Note that $\mathcal{F} = \bigcap_{i=1}^m \{x \mid f_i(x) \le 0\} = \bigcap_{i=1}^m L_0(f_i)$. By part (a), each $L_0(f_i)$ is convex, and by Problem 1.3(a), the intersection of convex sets is convex.
(d) Each fairness constraint $f_i(w) \le 0$ defines a convex sub-level set of the convex function f_i. The feasible region is $\mathcal{F} = \bigcap_{i=1}^m \{w : f_i(w) \le 0\}$, an intersection of convex sets. Even though individually each constraint may carve out a complex region, the intersection remains convex. This guarantees that any convex objective optimized over $\mathcal{F}$ yields a tractable convex program, and local optima are global optima.

Convex and Concave Functions — Characterizations

Exercise 2.1 [Easy]
Classify each of the following functions as convex, strictly convex, strongly convex, concave, or none. Justify each answer using the second-order condition where applicable.

(a) $f(x) = 3x^2 + 2x + 1$ on $\mathbb{R}$.
(b) $g(x) = -\log(x)$ on $\mathbb{R}_{++}$.
(c) $h(x_1, x_2) = x_1 x_2$ on $\mathbb{R}^2$.

Solution

(a) $f''(x) = 6 > 0$ for all x, so the Hessian (a scalar here) satisfies $f''(x) \geq 6 > 0$. In fact $f''(x) = 6 \geq 6 \cdot I$ (with $m = 6$), so f is **6-strongly convex**, hence also strictly convex and convex.
(b) $g'(x) = -1/x$, $g''(x) = 1/x^2 > 0$ for all $x > 0$. So g is **strictly convex** on $\mathbb{R}_{++}$. However, $g''(x) = 1/x^2 \to 0$ as $x \to \infty$, so there is no $m > 0$ with $g''(x) \geq m$ for all $x > 0$. Hence g is **not** strongly convex on $\mathbb{R}_{++}$.
Also, $-g(x) = \log(x)$ is concave (since log is concave), so g is convex, confirming the above.
(c) The Hessian of h is

$$H_h = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Eigenvalues: $\lambda = \pm 1$. Since H_h is indefinite (one positive, one negative eigenvalue), h is **neither convex nor concave**.

Exercise 2.2 [Easy]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and $g : \mathbb{R}^n \to \mathbb{R}$ be concave.
(a) Show that $h(x) = f(x) - g(x)$ is convex.
(b) Is $-f$ concave? Prove or disprove.
(c) Let $\alpha \geq 0$. Show that αf is convex.

Solution

(a) Since g is concave, $-g$ is convex. We show $h = f + (-g)$ is convex: for $\lambda \in [0, 1]$,

$$\begin{aligned} h(\lambda x + (1-\lambda)y) &= f(\lambda x + (1-\lambda)y) - g(\lambda x + (1-\lambda)y) \\ &\leq \lambda f(x) + (1-\lambda)f(y) - [\lambda g(x) + (1-\lambda)g(y)] \\ &= \lambda h(x) + (1-\lambda)h(y), \end{aligned}$$

where we used convexity of f ($\leq$ in first term) and concavity of g ($\geq$ in g term, which becomes $\leq$ after negation).
(b) Yes. For $\lambda \in [0, 1]$:

$$(-f)(\lambda x+(1-\lambda)y) = -f(\lambda x+(1-\lambda)y) \geq -(\lambda f(x)+(1-\lambda)f(y)) = \lambda(-f)(x)+(1-\lambda)(-f)(y).$$

The inequality follows from $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$ and negation flips the direction. This is the definition of concavity.
(c) For $\alpha \geq 0$ and $\lambda \in [0, 1]$:

$$(\alpha f)(\lambda x+(1-\lambda)y) = \alpha f(\lambda x+(1-\lambda)y) \leq \alpha[\lambda f(x)+(1-\lambda)f(y)] = \lambda(\alpha f)(x)+(1-\lambda)(\alpha f)(y),$$

since $\alpha \geq 0$ preserves the inequality direction.

Exercise 2.3 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable.
(a) State the first-order and second-order sufficient conditions for f to be convex.
(b) Using the second-order condition, prove that $f(x) = \|Ax - b\|_2^2$ is convex for any $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Under what condition on A is f strongly convex?
(c) In Ridge regression, the objective is

$$L(\theta) = \|X\theta - y\|_2^2 + \lambda\|\theta\|_2^2$$

for $\lambda > 0$. Show that L is 2λ-strongly convex regardless of X.

Solution

(a) *First-order condition:* f is convex if and only if for all $x, y \in \mathbb{R}^n$:

$$f(y) \geq f(x) + \nabla f(x)^T (y - x).$$

Second-order condition: f is convex if and only if for all $x \in \mathbb{R}^n$:

$$\nabla^2 f(x) \succeq 0 \quad \text{(positive semidefinite)}.$$

(b) Expand: $f(x) = (Ax - b)^T (Ax - b) = x^T A^T Ax - 2b^T Ax + b^T b$.
Gradient: $\nabla f(x) = 2A^T Ax - 2A^T b$.
Hessian: $\nabla^2 f(x) = 2A^T A$.
For any $v \in \mathbb{R}^n$: $v^T (2A^T A) v = 2\|Av\|_2^2 \geq 0$, so $\nabla^2 f \succeq 0$ and f is convex.
f is m-strongly convex iff $\nabla^2 f \succeq mI$, i.e., $2A^T A \succeq mI$. This holds with $m = 2\sigma_{\min}^2(A)$ where $\sigma_{\min}(A)$ is the smallest singular value of A. So f is strongly convex if and only if A has full column rank ($\sigma_{\min}(A) > 0$).
(c) The Hessian of L is:

$$\nabla^2 L(\theta) = 2X^T X + 2\lambda I.$$

For any $v \in \mathbb{R}^n$:

$$v^T \nabla^2 L(\theta)\, v = 2\|Xv\|_2^2 + 2\lambda \|v\|_2^2 \geq 2\lambda \|v\|_2^2.$$

Thus $\nabla^2 L(\theta) \succeq 2\lambda I$ for all θ, which means L is 2λ-strongly convex. This holds regardless of X because the regularization term contributes $2\lambda I$ to the Hessian, ensuring strong convexity even when X does not have full column rank.

Exercise 2.4 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and differentiable.

(a) Prove the first-order characterization: f is convex if and only if

$$f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall\, x, y \in \mathbb{R}^n.$$

(Hint: For the forward direction, use the definition of convexity with parameter $t \in (0, 1]$ and $z = x + t(y - x)$, then take $t \to 0^+$.)

(b) Use (a) to show that if f is convex and differentiable, then any critical point (where $\nabla f(x^*) = 0$) is a global minimizer.

Solution

(a)
($\Leftarrow$): Assume the first-order condition. For $\lambda \in [0,1]$, let $z = \lambda x + (1-\lambda)y$. Apply the condition twice:

$$f(x) \geq f(z) + \nabla f(z)^T(x-z),$$
$$f(y) \geq f(z) + \nabla f(z)^T(y-z).$$

Multiply the first by λ and the second by $(1-\lambda)$ and add:

$$\lambda f(x) + (1-\lambda)f(y) \geq f(z) + \nabla f(z)^T\big(\lambda(x-z) + (1-\lambda)(y-z)\big).$$

Note $\lambda(x-z) + (1-\lambda)(y-z) = \lambda x + (1-\lambda)y - z = 0$. So:

$$\lambda f(x) + (1-\lambda)f(y) \geq f(\lambda x + (1-\lambda)y),$$

which is the definition of convexity.
($\Rightarrow$): Assume f is convex. For $t \in (0,1]$:

$$f(x + t(y-x)) \leq (1-t)f(x) + tf(y).$$

Rearrange: $f(x+t(y-x)) - f(x) \leq t(f(y) - f(x))$. Divide by $t > 0$:

$$\frac{f(x+t(y-x)) - f(x)}{t} \leq f(y) - f(x).$$

Taking $t \to 0^+$, the left side becomes $\nabla f(x)^T(y-x)$ by the definition of the directional derivative. Therefore:

$$\nabla f(x)^T(y-x) \leq f(y) - f(x) \quad \Longrightarrow \quad f(y) \geq f(x) + \nabla f(x)^T(y-x).$$

(b) If $\nabla f(x^*) = 0$, then by the first-order condition:

$$f(y) \geq f(x^*) + 0^T(y - x^*) = f(x^*), \quad \forall\, y \in \mathbb{R}^n.$$

Thus x^* is a global minimizer.

Exercise 2.5 [Hard]
(a) Let $f : \mathbb{R}^n \to \mathbb{R}$ be m-strongly convex ($m > 0$). Prove that f has a unique global minimizer.
(Hint: Use the strong convexity inequality $f(y) \geq f(x) + \nabla f(x)^T(y-x) + \frac{m}{2}\|y-x\|^2$ to show existence of a lower bound, and then prove uniqueness by contradiction.)

(b) Let f be strictly convex on a convex set $C \subseteq \mathbb{R}^n$. Show that f has at most one minimizer over C. Give an example showing that a minimizer need not exist (i.e., strict convexity does not guarantee existence without further assumptions).

(c) Consider the negative log-likelihood in logistic regression:

$$L(w) = \sum_{i=1}^{N} \Big[\log\big(1 + \exp(w^T x_i)\big) - y_i\, w^T x_i\Big],$$

where $y_i \in \{0,1\}$ and $x_i \in \mathbb{R}^d$. Show that L is convex. Is it strongly convex in general? What common regularization technique makes the objective strongly convex, and why?

Solution

(a) *Existence:* Since f is m-strongly convex, $f(x) \geq f(x_0) + \nabla f(x_0)^T(x - x_0) + \frac{m}{2}\|x - x_0\|^2$ for any fixed x_0. As $\|x\| \to \infty$, the quadratic term dominates, so $f(x) \to \infty$ (coercive). By continuity and coercivity, a minimizer x^* exists.
Uniqueness: Suppose x^* and $\hat{x}$ are both global minimizers with $f(x^*) = f(\hat{x}) = f^*$ and $x^* \neq \hat{x}$. By strong convexity with $\lambda = 1/2$:

$$f\left(\frac{x^* + \hat{x}}{2}\right) \leq \frac{1}{2}f(x^*) + \frac{1}{2}f(\hat{x}) - \frac{m}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}\|x^* - \hat{x}\|^2 = f^* - \frac{m}{8}\|x^* - \hat{x}\|^2 < f^*,$$

contradicting optimality. Therefore the minimizer is unique.
(b) *At most one minimizer:* Suppose x^* and $\hat{x}$ are both minimizers over C with $x^* \neq \hat{x}$. Since C is convex, $z = \frac{1}{2}x^* + \frac{1}{2}\hat{x} \in C$. By strict convexity:

$$f(z) < \frac{1}{2}f(x^*) + \frac{1}{2}f(\hat{x}) = f^*,$$

contradicting that f^* is the minimum over C.
Existence not guaranteed: Take $f(x) = e^x$ on $C = \mathbb{R}$. Then f is strictly convex ($f''(x) = e^x > 0$), but $\inf_{x \in \mathbb{R}} e^x = 0$ is never attained.
(c) Each summand has the form $\ell_i(w) = \log(1 + e^{w^T x_i}) - y_i w^T x_i$.
The Hessian of $\log(1 + e^z)$ with respect to z is $\sigma(z)(1 - \sigma(z))$ where $\sigma(z) = \frac{e^z}{1+e^z}$. By the chain rule:

$$\nabla_w^2 \ell_i(w) = \sigma(w^T x_i)(1 - \sigma(w^T x_i))\, x_i x_i^T \succeq 0,$$

since $\sigma(1 - \sigma) > 0$ and $x_i x_i^T \succeq 0$. Thus $\nabla^2 L(w) = \sum_i \sigma_i(1 - \sigma_i) x_i x_i^T \succeq 0$, so L is convex.
L is **not strongly convex** in general: if the data $\{x_i\}$ do not span $\mathbb{R}^d$ (e.g., more features than samples), then $X^T X$ is singular and the Hessian has zero eigenvalues.
L_2 **regularization** (adding $\frac{\lambda}{2}\|w\|^2$ with $\lambda > 0$) makes the objective strongly convex. The regularized Hessian becomes $\nabla^2 L(w) + \lambda I \succeq \lambda I \succ 0$, guaranteeing λ-strong convexity and thus a unique global minimum.

Exercise 2.6 [Hard]
Let $f_1, \ldots, f_m : \mathbb{R}^n \to \mathbb{R}$ be convex functions, and let $A \in \mathbb{R}^{n \times p}$, $b \in \mathbb{R}^n$.

(a) Prove that $g(x) = \sum_{i=1}^m \alpha_i f_i(x)$ is convex for any $\alpha_i \geq 0$ (non-negative weighted sum preserves convexity).

(b) Prove that $h(y) = f_1(Ay + b)$ is convex in y (composition with affine mapping preserves convexity).

(c) Prove that $\phi(x) = \max_{i=1,\ldots,m} f_i(x)$ is convex (pointwise maximum preserves convexity).

(d) Give a counterexample showing that the composition $f_1 \circ f_2$ of two convex functions is not necessarily convex. What additional condition on f_1 makes it work?

Solution

(a) For $\lambda \in [0,1]$:

$$g(\lambda x + (1-\lambda)y) = \sum_i \alpha_i f_i(\lambda x + (1-\lambda)y) \leq \sum_i \alpha_i [\lambda f_i(x) + (1-\lambda) f_i(y)]$$

$$= \lambda \sum_i \alpha_i f_i(x) + (1-\lambda) \sum_i \alpha_i f_i(y) = \lambda g(x) + (1-\lambda) g(y).$$

The inequality uses convexity of each f_i and $\alpha_i \geq 0$.
(b) For $\lambda \in [0,1]$:

$$\begin{aligned} h(\lambda y_1 + (1-\lambda)y_2) &= f_1(A(\lambda y_1 + (1-\lambda)y_2) + b) \\ &= f_1(\lambda(Ay_1 + b) + (1-\lambda)(Ay_2 + b)) \\ &\leq \lambda f_1(Ay_1 + b) + (1-\lambda) f_1(Ay_2 + b) \\ &= \lambda h(y_1) + (1-\lambda) h(y_2). \end{aligned}$$

The key step is that an affine map distributes over convex combinations, so we can apply convexity of f_1 directly.
(c) For $\lambda \in [0,1]$:

$$\phi(\lambda x + (1-\lambda)y) = \max_i f_i(\lambda x + (1-\lambda)y) \leq \max_i [\lambda f_i(x) + (1-\lambda) f_i(y)]$$

$$\leq \lambda \max_i f_i(x) + (1-\lambda) \max_i f_i(y) = \lambda \phi(x) + (1-\lambda)\phi(y).$$

The second inequality uses $\max(a_i + b_i) \leq \max(a_i) + \max(b_i)$ for non-negative weighted terms.
(d) *Counterexample:* Let $f_1(t) = t^2$ (convex) and $f_2(x) = -x$ (convex, since it is affine). Then $(f_1 \circ f_2)(x) = (-x)^2 = x^2$ — this happens to be convex. A better counterexample: $f_1(t) = (t-1)^2$ and $f_2(x) = x^2$. Both are convex. But $(f_1 \circ f_2)(x) = (x^2-1)^2 = x^4 - 2x^2 + 1$. The second derivative is $12x^2 - 4$, which is negative for $|x| < 1/\sqrt{3}$, so $f_1 \circ f_2$ is not convex. *Sufficient condition:* If f_1 is convex and **non-decreasing**, and f_2 is convex, then $f_1 \circ f_2$ is convex. In the counterexample, $f_1(t) = (t-1)^2$ is not non-decreasing.

Jensen's Inequality and Applications

Exercise 3.1 [Easy]
Jensen's inequality states: if f is convex and X is a random variable with $\mathbb{E}[|X|] < \infty$, then

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

(a) Using $f(t) = t^2$, derive the inequality $\mathrm{Var}(X) \geq 0$ from Jensen's inequality.
(b) Using $f(t) = -\log(t)$ for $t > 0$, show that for positive values $a_1, \ldots, a_n$:

$$\frac{1}{n} \sum_{i=1}^{n} a_i \geq \left(\prod_{i=1}^{n} a_i \right)^{1/n}$$

(the AM-GM inequality).

Solution

(a) Since $f(t) = t^2$ is convex, Jensen gives:

$$(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2].$$

Therefore $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \geq 0$.
(b) Since $f(t) = -\log(t)$ is convex on $\mathbb{R}_{++}$, applying Jensen with X uniform on $\{a_1, \ldots, a_n\}$ (i.e., $\mathbb{E}[g(X)] = \frac{1}{n}\sum g(a_i)$):

$$-\log\left(\frac{1}{n}\sum_{i=1}^{n} a_i\right) \leq \frac{1}{n}\sum_{i=1}^{n}(-\log a_i) = -\frac{1}{n}\log\prod_{i=1}^{n} a_i = -\log\left(\prod_{i=1}^{n} a_i\right)^{1/n}.$$

Multiply both sides by -1 (flipping the inequality):

$$\log\left(\frac{1}{n}\sum a_i\right) \geq \log\left(\prod a_i\right)^{1/n}.$$

Since log is strictly increasing, this gives $\frac{1}{n}\sum a_i \geq (\prod a_i)^{1/n}$.

Exercise 3.2 [Medium]
Let X be a random variable taking values in a convex set $C \subseteq \mathbb{R}^n$, and let $f : C \to \mathbb{R}$ be a **concave** function.

(a) State and prove Jensen's inequality for concave functions.

(b) Show that for any non-negative random variable X with finite mean:

$$\mathbb{E}[\sqrt{X}] \leq \sqrt{\mathbb{E}[X]}.$$

(c) In Bayesian machine learning, the *marginal log-likelihood* is $\log p(y) = \log \int p(y \mid w)p(w)\,dw$. Use Jensen's inequality applied to log (which is concave) to derive the *Evidence Lower Bound* (ELBO): for any distribution $q(w)$,

$$\log p(y) \geq \mathbb{E}_{q(w)}\left[\log \frac{p(y \mid w)p(w)}{q(w)}\right].$$

Solution

(a) If f is concave, Jensen's inequality reverses: $f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$.
Proof: Since f is concave, $-f$ is convex. By Jensen's inequality for convex functions:

$$(-f)(\mathbb{E}[X]) \leq \mathbb{E}[(-f)(X)] \quad \Longrightarrow \quad -f(\mathbb{E}[X]) \leq -\mathbb{E}[f(X)].$$

Multiplying by -1: $f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$.
(b) $f(t) = \sqrt{t}$ is concave on $[0, \infty)$ (since $f''(t) = -\frac{1}{4}t^{-3/2} < 0$ for $t > 0$). By Jensen for concave functions:

$$\mathbb{E}[\sqrt{X}] = \mathbb{E}[f(X)] \leq f(\mathbb{E}[X]) = \sqrt{\mathbb{E}[X]}.$$

(c) Write:

$$\log p(y) = \log \int p(y \mid w)p(w)\, dw = \log \int \frac{p(y \mid w)p(w)}{q(w)} q(w)\, dw = \log \mathbb{E}_{q(w)}\left[\frac{p(y \mid w)p(w)}{q(w)}\right].$$

Since log is concave, Jensen's inequality gives:

$$\log \mathbb{E}_q\left[\frac{p(y|w)p(w)}{q(w)}\right] \geq \mathbb{E}_q\left[\log \frac{p(y|w)p(w)}{q(w)}\right].$$

Therefore: $\log p(y) \geq \mathbb{E}_{q(w)}\left[\log \frac{p(y|w)p(w)}{q(w)}\right]$, which is the ELBO.

Exercise 3.3 [Hard]
(a) Let p and q be probability distributions over a finite set $\{1, \ldots, n\}$. The KL divergence is defined as

$$D_{\mathrm{KL}}(q\|p) = \sum_{i=1}^{n} q_i \log \frac{q_i}{p_i}.$$

Using Jensen's inequality applied to the convex function $f(t) = -\log(t)$, prove that $D_{\mathrm{KL}}(q\|p) \geq 0$ with equality if and only if $q = p$.

(b) Now consider the optimization problem

$$\min_q \; D_{\mathrm{KL}}(q\|p), \quad \text{s.t.} \quad q_i \geq 0, \; \sum_{i=1}^{n} q_i = 1.$$

Using part (a), what is the minimizer? Relate this to the entropy maximization problem

$$\max_q \; H(q) = -\sum_{i=1}^{n} q_i \log q_i, \quad \text{s.t.} \quad q_i \geq 0, \; \sum_{i=1}^{n} q_i = 1,$$

when p is the uniform distribution $p_i = 1/n$.

Solution

(a) We have:

$$D_{\mathrm{KL}}(q\|p) = \sum_i q_i \log \frac{q_i}{p_i} = -\sum_i q_i \log \frac{p_i}{q_i} = \mathbb{E}_q\left[-\log \frac{p_i}{q_i}\right].$$

Apply Jensen's inequality to the convex function $f(t) = -\log(t)$, with random variable $T = p_i/q_i$ under distribution q:

$$\mathbb{E}_q\left[-\log \frac{p_i}{q_i}\right] \geq -\log \mathbb{E}_q\left[\frac{p_i}{q_i}\right] = -\log \sum_i q_i \cdot \frac{p_i}{q_i} = -\log \sum_i p_i = -\log 1 = 0.$$

Thus $D_{\mathrm{KL}}(q\|p) \geq 0$.
Equality: Since $-\log$ is strictly convex, Jensen's inequality is tight if and only if p_i/q_i is constant q-a.s., i.e., $p_i/q_i = c$ for all i with $q_i > 0$. Since both are probability distributions summing to 1, $c = 1$, so $q = p$.
(b) By part (a), $D_{\mathrm{KL}}(q\|p) \geq 0$ with equality iff $q = p$. So the minimizer is $q^* = p$.
When $p_i = 1/n$ (uniform), expand:

$$D_{\mathrm{KL}}(q\|p) = \sum_i q_i \log \frac{q_i}{1/n} = \sum_i q_i \log q_i + \sum_i q_i \log n = -H(q) + \log n.$$

So $\min_q D_{\mathrm{KL}}(q\|p) = \min_q[-H(q) + \log n] = -\max_q H(q) + \log n$.
Since the KL minimizer is $q^* = p = (1/n, \ldots, 1/n)$, this is also the entropy maximizer, with $H(q^*) = \log n$.
Thus, maximizing entropy over the simplex is equivalent to minimizing KL divergence to the uniform distribution.

Epigraph, Subdifferential, Normal Cone, and Extreme Points

Exercise 4.1 [Medium]
(a) The *epigraph* of a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is defined as

$$\operatorname{epi} f = \{(x, t) \in \mathbb{R}^{n+1} \mid f(x) \leq t\}.$$

Prove that f is convex if and only if $\operatorname{epi} f$ is a convex set.

(b) Consider the *extended-value* function $f : \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ defined by $f(x) = -\sqrt{x}$ for $x \geq 0$ and $f(x) = +\infty$ for $x < 0$. Sketch $\operatorname{epi} f$ and determine whether f is convex.

Solution

(a) ($\Rightarrow$): Assume f is convex. Let $(x_1, t_1), (x_2, t_2) \in \operatorname{epi} f$, so $f(x_1) \leq t_1$ and $f(x_2) \leq t_2$. For $\lambda \in [0, 1]$:

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2) \leq \lambda t_1 + (1-\lambda) t_2.$$

Thus $(\lambda x_1 + (1-\lambda)x_2,\ \lambda t_1 + (1-\lambda)t_2) \in \operatorname{epi} f$, so $\operatorname{epi} f$ is convex.
($\Leftarrow$): Assume $\operatorname{epi} f$ is convex. For $x_1, x_2 \in \operatorname{dom} f$ and $\lambda \in [0, 1]$, the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ are in $\operatorname{epi} f$. By convexity of $\operatorname{epi} f$:

$$(\lambda x_1 + (1-\lambda)x_2,\ \lambda f(x_1) + (1-\lambda) f(x_2)) \in \operatorname{epi} f.$$

By definition of the epigraph:

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2).$$

(b) The effective domain is $[0, \infty)$. On this domain, $f(x) = -\sqrt{x}$ has $f''(x) = \frac{1}{4}x^{-3/2} > 0$ for $x > 0$, so f is **not convex** on $[0, \infty)$—wait, $f''(x) > 0$ means f *is* convex on $(0, \infty)$. Let us verify: $f(x) = -\sqrt{x}$, $f'(x) = -\frac{1}{2\sqrt{x}}$, $f''(x) = \frac{1}{4x^{3/2}} > 0$. So f is indeed convex on $(0, \infty)$, and it extends to a convex function on $[0, \infty)$ by continuity. With $f(x) = +\infty$ for $x < 0$, the extended function is convex.
The epigraph $\operatorname{epi} f = \{(x, t) : x \geq 0,\ t \geq -\sqrt{x}\}$ is the region above the curve $t = -\sqrt{x}$ for $x \geq 0$. This is a convex set (it includes the entire half-plane $x < 0$ is excluded since those points have $f = +\infty$ and no finite t works). So $\operatorname{epi} f$ is convex, consistent with f being convex.

Exercise 4.2 [Medium]
(a) Compute the subdifferential $\partial f(x)$ of $f(x) = |x|$ at $x = 0$, $x = 2$, and $x = -3$.
(b) Let $f(x) = \max(0, 1 - x)$ (the hinge loss used in SVMs). Compute $\partial f(x)$ for all $x \in \mathbb{R}$.
(c) State and verify the optimality condition $0 \in \partial f(x^*)$ for the function $f(x) = |x-3| + 2|x|$. Find all minimizers.

Solution

(a) A subgradient v at x satisfies $|z| \geq |x| + v(z - x)$ for all $z \in \mathbb{R}$.
At $x = 0$*:* $|z| \geq 0 + vz$ for all z, so $v \leq |z|/z$ for $z > 0$ (giving $v \leq 1$) and $v \geq |z|/z$ for $z < 0$ (giving $v \geq -1$). Thus $\partial f(0) = [-1, 1]$.
At $x = 2$*:* f is differentiable with $f'(2) = 1$, so $\partial f(2) = \{1\}$.
At $x = -3$*:* f is differentiable with $f'(-3) = -1$, so $\partial f(-3) = \{-1\}$.
(b) $f(x) = \max(0, 1 - x)$. Three cases:
$x > 1$*:* $f(x) = 0$, differentiable, $\partial f(x) = \{0\}$.
$x < 1$*:* $f(x) = 1 - x$, differentiable, $\partial f(x) = \{-1\}$.
$x = 1$*:* $f(1) = 0$. Here $f = \max(f_1, f_2)$ with $f_1 = 0$, $f_2 = 1 - x$, both active. By the subdifferential rule for max: $\partial f(1) = \operatorname{conv}(\{0\} \cup \{-1\}) = [-1, 0]$.
(c) $f(x) = |x - 3| + 2|x|$. The subdifferential is:

$$\partial f(x) = \partial|x - 3| + 2\,\partial|x| = \operatorname{sign}(x - 3) + 2\operatorname{sign}(x),$$

where $\operatorname{sign}(0) = [-1, 1]$.
Check $x = 0$*:* $\partial f(0) = \{-1\} + 2[-1, 1] = \{-1\} + [-2, 2] = [-3, 1]$. Since $0 \in [-3, 1]$, $x = 0$ is a minimizer. ✓
Check $x = 3$*:* $\partial f(3) = [-1, 1] + \{2\} = [1, 3]$. Since $0 \notin [1, 3]$, $x = 3$ is not a minimizer.
For $0 < x < 3$*:* $\partial f(x) = \{-1\} + \{2\} = \{1\}$. Since $0 \notin \{1\}$, no minimizer here.
For $x < 0$*:* $\partial f(x) = \{-1\} + \{-2\} = \{-3\}$. Not a minimizer.
For $x > 3$*:* $\partial f(x) = \{1\} + \{2\} = \{3\}$. Not a minimizer.
The unique minimizer is $x^* = 0$ with $f(0) = 3$.

Exercise 4.3 [Medium]
(a) Let $C = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq r\}$ be the closed ball of radius r. Compute the normal cone $N_C(x)$ for:

- $x \in \operatorname{int}(C)$ (i.e., $\|x\|_2 < r$),
- $x \in \partial C$ (i.e., $\|x\|_2 = r$).

(b) A point x in a convex set C is an *extreme point* if x cannot be written as $x = \lambda y + (1-\lambda)z$ for $y, z \in C$ with $y \neq z$ and $\lambda \in (0, 1)$. Find all extreme points of:

- The simplex $\Delta = \{x \in \mathbb{R}^n \mid x_i \geq 0,\ \sum x_i = 1\}$.
- The ℓ_∞-ball $B_\infty = \{x \in \mathbb{R}^2 \mid \|x\|_\infty \leq 1\}$.

Solution

(a) The normal cone of C at x is $N_C(x) = \{s \in \mathbb{R}^n \mid \langle s, y - x\rangle \leq 0,\ \forall y \in C\}$.
Interior ($\|x\|_2 < r$): Since $x \in \text{int}(C)$, there exists $\epsilon > 0$ such that $B(x, \epsilon) \subset C$. For any $s \neq 0$, choosing $y = x + \frac{\epsilon}{2}\frac{s}{\|s\|}$ gives $\langle s, y - x\rangle = \frac{\epsilon}{2}\|s\| > 0$, violating the normal cone condition. Thus $N_C(x) = \{0\}$.
Boundary ($\|x\|_2 = r$): For $y \in C$, we need $\langle s, y - x\rangle \leq 0$. The constraint $\|y\|_2 \leq r = \|x\|_2$ implies (after some calculation using Cauchy-Schwarz) that $s = \alpha x$ for some $\alpha \geq 0$ works: $\langle \alpha x, y - x\rangle = \alpha(\langle x, y\rangle - r^2) \leq \alpha(\|x\|\|y\| - r^2) \leq 0$. Conversely, any s in the normal cone must point in the direction of x (outward normal). Thus $N_C(x) = \{\alpha x \mid \alpha \geq 0\}$.
(b)
Simplex Δ: The extreme points are the standard basis vectors $e_1, \ldots, e_n$. If $x = \lambda y + (1-\lambda)z$ with $y, z \in \Delta$ and $\lambda \in (0, 1)$, then if $x = e_i$ (so $x_i = 1$ and $x_j = 0$ for $j \neq i$), then $0 = \lambda y_j + (1 - \lambda)z_j$ with $y_j, z_j \geq 0$ forces $y_j = z_j = 0$ for all $j \neq i$, hence $y = z = e_i$. Conversely, any $x \in \Delta$ with more than one non-zero component can be decomposed: if $x_i, x_j > 0$ with $i \neq j$, perturb by $\pm\epsilon(e_i - e_j)$ to get two distinct points in Δ averaging to x.
ℓ_∞-ball B_∞ in $\mathbb{R}^2$: $B_\infty = [-1, 1]^2$ is a square. The extreme points are the four corners: $(\pm 1, \pm 1)$. Any point on an edge (but not a corner) is a convex combination of the two adjacent corners.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
Consider the *log-sum-exp* function $f : \mathbb{R}^n \to \mathbb{R}$ defined as

$$f(x) = \log\left(\sum_{i=1}^{n} e^{x_i}\right).$$

(a) Prove that f is convex using Jensen's inequality.
(Hint: Interpret f as $\log \mathbb{E}[e^X]$ for an appropriate discrete random variable.)
(b) Show that $\max_i x_i \leq f(x) \leq \max_i x_i + \log n$. (This shows that log-sum-exp is a smooth approximation to the max function.)
(c) Compute the gradient $\nabla f(x)$ and the Hessian $\nabla^2 f(x)$. Show that $\nabla^2 f(x) \succeq 0$, giving an alternative proof of convexity.
(Hint: The Hessian can be written as $\text{diag}(s) - ss^T$ where $s = \text{softmax}(x)$. Use Cauchy-Schwarz to show $v^T(\text{diag}(s) - ss^T)v \geq 0$.)

Solution

(a) Write $f(x) = \log\left(n \cdot \frac{1}{n}\sum_{i=1}^n e^{x_i}\right) = \log n + \log\left(\mathbb{E}[e^X]\right)$, where X is uniform on $\{x_1, \ldots, x_n\}$. Since e^t is convex, Jensen gives $\mathbb{E}[e^X] \geq e^{\mathbb{E}[X]}$, which shows $f(x) \geq \mathbb{E}[X] + \log n$.
Alternatively and more directly: for $\lambda \in [0,1]$, we show $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$:

$$\sum_i e^{\lambda x_i + (1-\lambda)y_i} = \sum_i (e^{x_i})^\lambda (e^{y_i})^{1-\lambda} \leq \sum_i [\lambda e^{x_i} + (1-\lambda)e^{y_i}] = \lambda \sum_i e^{x_i} + (1-\lambda)\sum_i e^{y_i},$$

where we used the AM-GM inequality $a^\lambda b^{1-\lambda} \leq \lambda a + (1-\lambda)b$ for $a, b > 0$. Taking log (which is monotone):

$$f(\lambda x + (1-\lambda)y) \leq \log(\lambda e^{f(x)} + (1-\lambda)e^{f(y)}) \leq \lambda f(x) + (1-\lambda)f(y),$$

where the last step uses concavity of log.
(b) *Lower bound:* Let $k = \arg\max_i x_i$. Then $\sum_i e^{x_i} \geq e^{x_k}$, so $f(x) \geq x_k = \max_i x_i$.
Upper bound: $\sum_i e^{x_i} \leq \sum_i e^{\max_j x_j} = n \cdot e^{\max_i x_i}$, so $f(x) \leq \max_i x_i + \log n$.
(c) Let $S = \sum_i e^{x_i}$ and $s_i = e^{x_i}/S$ (softmax).
Gradient: $\frac{\partial f}{\partial x_i} = \frac{e^{x_i}}{S} = s_i$, so $\nabla f(x) = s = \text{softmax}(x)$.
Hessian: $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial s_i}{\partial x_j} = s_i(\delta_{ij} - s_j)$. In matrix form: $\nabla^2 f = \text{diag}(s) - ss^T$.
Positive semidefiniteness: For any $v \in \mathbb{R}^n$:

$$v^T(\text{diag}(s) - ss^T)v = \sum_i s_i v_i^2 - \left(\sum_i s_i v_i\right)^2.$$

Since $s_i > 0$ and $\sum s_i = 1$, this is $\mathbb{E}[V^2] - (\mathbb{E}[V])^2 = \text{Var}(V) \geq 0$, where V takes value v_i with probability s_i. This is exactly the Cauchy-Schwarz / Jensen inequality, confirming $\nabla^2 f \succeq 0$.

Exercise 5.2 [Hard]
Consider the following optimization problem from statistical learning. Let $p \in \mathbb{R}^n$ with $p_i > 0$ and $\sum p_i = 1$ be a fixed probability distribution. Solve:

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^n x_i \left(\log x_i - \log p_i\right), \quad \text{s.t.} \quad x_i \geq 0, \ \sum_{i=1}^n x_i = 1.$$

(a) Show that the objective function $\phi(x) = \sum_i x_i(\log x_i - \log p_i)$ is strictly convex on the simplex $\{x \mid x_i > 0, \sum x_i = 1\}$.

(b) Write the Lagrangian with multiplier μ for the equality constraint and multipliers λ_i for the inequality constraints $-x_i \leq 0$.

(c) Apply the KKT conditions to find the optimal solution x^*.

(d) Interpret the result: what does it say about when the KL divergence $D_{\text{KL}}(x\|p)$ is minimized?

Solution

(a) The Hessian of ϕ restricted to the simplex: $\frac{\partial^2 \phi}{\partial x_i^2} = \frac{1}{x_i}$ and $\frac{\partial^2 \phi}{\partial x_i \partial x_j} = 0$ for $i \neq j$. So $\nabla^2 \phi = \operatorname{diag}(1/x_1, \ldots, 1/x_n)$. For $x_i > 0$, this is positive definite (all diagonal entries positive), so ϕ is strictly convex on the open simplex.
(b) Lagrangian:

$$\mathcal{L}(x, \mu, \lambda) = \sum_{i=1}^{n} x_i(\log x_i - \log p_i) + \mu\Big(\sum_{i=1}^{n} x_i - 1\Big) + \sum_{i=1}^{n} \lambda_i(-x_i).$$

(c) KKT conditions:
1. *Stationarity:* $\frac{\partial \mathcal{L}}{\partial x_i} = \log x_i + 1 - \log p_i + \mu - \lambda_i = 0$ for all i.
2. *Primal feasibility:* $\sum x_i = 1$, $x_i \geq 0$.
3. *Dual feasibility:* $\lambda_i \geq 0$.
4. *Complementary slackness:* $\lambda_i x_i = 0$ for all i.

From stationarity: $\log x_i = \log p_i - 1 - \mu + \lambda_i$, so $x_i = p_i \cdot e^{-1-\mu+\lambda_i}$.
If $x_i > 0$, complementary slackness gives $\lambda_i = 0$, so $x_i = p_i \cdot e^{-1-\mu}$.
The equality constraint $\sum x_i = 1$ gives $e^{-1-\mu} \sum_{i:x_i>0} p_i = 1$.
If all $x_i > 0$: $e^{-1-\mu} \cdot 1 = 1$ (since $\sum p_i = 1$), so $e^{-1-\mu} = 1$ and $x_i = p_i$ for all i.
We verify: $\lambda_i = 0 \geq 0$ ✓, $x_i = p_i > 0$ ✓. All KKT conditions satisfied.
The optimal solution is $x^* = p$.
(d) The objective $\phi(x) = D_{\mathrm{KL}}(x\|p)$ is the KL divergence from x to p. The result confirms that $D_{\mathrm{KL}}(x\|p)$ is minimized when $x = p$, with minimum value $D_{\mathrm{KL}}(p\|p) = 0$. This is consistent with Gibbs' inequality ($D_{\mathrm{KL}} \geq 0$).

Exercise 5.3 [Hard]
Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive semidefinite and $b \in \mathbb{R}^n$.

(a) Consider the set

$$\mathcal{E} = \{x \in \mathbb{R}^n \mid x^T A x + b^T x \leq 1\}.$$

Prove that $\mathcal{E}$ is a convex set.

(b) Let $f : \mathbb{R}^n \to \mathbb{R}$ be a strongly convex function with parameter $m > 0$. Prove that for any minimizer x^* and any $x \in \mathbb{R}^n$:

$$f(x) - f(x^*) \geq \frac{m}{2}\|x - x^*\|^2.$$

(c) A data scientist is training a model by minimizing $L(w) + \frac{\lambda}{2}\|w\|^2$ where L is convex but not strongly convex, and $\lambda > 0$. The training algorithm reaches a point $\hat{w}$ with $f(\hat{w}) - f(w^*) \leq \epsilon$. Using (b), derive an upper bound on $\|\hat{w} - w^*\|$ in terms of ϵ and λ.

Solution

(a) Define $g(x) = x^T Ax + b^T x$. The gradient is $\nabla g(x) = 2Ax + b$ and the Hessian is $\nabla^2 g(x) = 2A \succeq 0$ (since A is PSD). So g is convex. The set $\mathcal{E} = \{x : g(x) \leq 1\}$ is the sub-level set $L_1(g)$ of a convex function, hence convex (by the general result that sub-level sets of convex functions are convex).

(b) Since f is m-strongly convex and x^* is the minimizer, by the first-order strong convexity condition:

$$f(x) \geq f(x^*) + \nabla f(x^*)^T (x - x^*) + \frac{m}{2}\|x - x^*\|^2.$$

Since x^* is the minimizer and f is differentiable (strongly convex functions on $\mathbb{R}^n$ are), $\nabla f(x^*) = 0$. Therefore:

$$f(x) \geq f(x^*) + \frac{m}{2}\|x - x^*\|^2,$$

which gives $f(x) - f(x^*) \geq \frac{m}{2}\|x - x^*\|^2$.

(c) The full objective $f(w) = L(w) + \frac{\lambda}{2}\|w\|^2$ is λ-strongly convex (since L is convex and the Hessian of the regularizer is λI). By part (b):

$$\epsilon \geq f(\hat{w}) - f(w^*) \geq \frac{\lambda}{2}\|\hat{w} - w^*\|^2.$$

Rearranging:

$$\|\hat{w} - w^*\| \leq \sqrt{\frac{2\epsilon}{\lambda}}.$$

This shows that stronger regularization (λ larger) gives a tighter bound on how close the approximate solution must be to the optimum.